



CS304-OOP
(Solved Macq's)
**LECTURE FROM
(23 to 45)**



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1. Function func1 of class1 is_____ in class2.
 - ▶ Public
 - ▶ **Protected**
 - ▶ Private
 - ▶ None of the given
2. The default inheritance mode is,
 - a) Public inheritance
 - b) Protected inheritance
 - c) Private inheritance**
 - d) None of the given
3. A template provides a convenient way to make a family of
 - a) Variables and data members
 - b) Functions and classes**
 - c) Classes and exceptions
 - d) Programs and algorithms
4. Then int member i of base class is accessible in class,
 - a) derived 1 only
 - b) derived 2 only**
 - c) both derived 1 derived 2
 - d) none of the given options
5. In case of public inheritance, protected members of base class will be_____in derived class?
 - a) Private
 - b) Public
 - c) Protected**
 - d) hidden
6. By default assignment operator (=) performs,
 - a) Shallow copy**
 - b) Deep copy
 - c) Both of these
 - d) None of these
7. In protected inheritance the public members of base class become _____in derived class.
 - a) Public
 - b) Private
 - c) Protected**
 - d) None of the given
8. We can access public members of the class from outside the class using_____operator with its object pointer
 - a) ->

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- b) .
c) &
d) @
9. Compiler can automatically generates the following constructor/s for a class,
a) Default constructor
b) Copy constructor
c) Both of these
d) None of these
10. Friend's functions of class are _____members of that class.
a) Public
b) Private
c) Protected
d) None of the given
11. Methodologies to the development of reusable software relate to _____.
a) Structure programming
b) Procedural programming
c) Generic programming
d) None of the given
12. Function overriding is done in context of,
a) Single class
b) Single derived class
c) Single base class
d) Derived and base classes
13. Which of the following is not type of inheritance in c++?
a) Public
b) Protected
c) Restricted
d) Private
14. If there is a pointer ,p, to objects of a base class and it contains the address of an object of a derived class and both classes contain a virtual member function, ding(),then the statement p->ding(); will cause the version of ding in the _____class to be executed.
a) Base
b) Derived
c) Virtual
d) Implemented
15. Target class of a _____function call is determined t run time.
a) Instance
b) Virtual
c) Operator
d) None of the given
16. In c++ dynamic binding and polymorphism will be achieved when the member function will be _____

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- a) Private
 - b) Public
 - c) **Virtual**
 - d) Inline
17. To make a pure virtual, we need to give ____ after() of this function.
- a) **=0;**
 - b) =1;
 - c) Null;
 - d) None of the given
18. If we have not given any constructor for the class, compiler generates which of the following constructors?
- a) Explicit Default Constructor
 - b) **Implicit Default Constructor**
 - c) Explicit Parameterized Constructor
 - d) Implicit Parameterized Constructor
19. In c++ by default access of classes is ____.
- a) **Private**
 - b) Protected
 - c) Public
 - d) None of the given
20. In public inheritance the public members of base class become ____ in derived class
- a) Private
 - b) Protected
 - c) **Public**
 - d) None of the given
21. A class hierarchy
- a) **Shows the same relationships as a organization chart**
 - b) Describe "has a" relationship.
 - c) Describe "is a kind of" relationship.
 - d) Shows the same relationships as a family tree.
22. In private Inheritance the public members of the base class become ____ in derived class
- a) Public
 - b) **private**
 - c) Protected
 - d) None of the given option
23. In the line No.2, which constructor of student class will be
- a) Default constructor of student class
 - b) **Copy constructor of student class**
 - c) Both Default and Copy constructor of student class
 - d) No constructor will be called
24. Which of the following is Not casting operator in c++ standard?
- a) Static-cast

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- b) **Var-cast**
 - c) Dynamic-cast
 - d) Reinterpret-cast
25. Which of the following type of inheritance is used to model “Implemented in terms of” relationship?
- a) Public
 - b) **Private**
 - c) Protected
 - d) Confidential
26. When we want to implement one class in terms of another class then we use, a) **Public inheritance**
- b) Protected inheritance
 - c) Private inheritance
 - d) None of these options
27. Public inheritance represents;
- a) **“IS A” relationship**
 - b) “Has A” relationship
 - c) “IS Special kind of” relationship
 - d) None of these option
28. Suppose we have defined derived class copy constructor but have not defined base class copy constructor then compiler will,
- a) Use base class default constructor
 - b) **Generate base class copy constructor itself**
 - c) Use base class assignment operator
 - d) Nine of these option is correct
29. Child class can call constructor of its, a) **Direct base class**
- b) Indirect base class
 - c) Both direct and indirect base classes
 - d) None of these
30. We can have _____ type of member functions in a class.
- a) Public
 - b) Private
 - c) Protected
 - d) **All of these options**
31. Compiler generated copy constructor performs _____
- a) **Shallow copy**
 - b) Deep copy
 - c) Both Shallow and Deep copy
 - d) None of these options
32. In c++ generic programming is done using _____
- a) Procedures

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- b) Packages
 - c) **Templates**
 - d) None of the given
33. Target class of a ____ function call is determined at run time.
- a) Instance
 - b) **Virtual**
 - c) Operator
 - d) None of the given
34. Suppose person is a user defined class. In statement "Person*pPtr", static type of pPtr is ----.
- a) pPtr
 - b) **pointer**
 - c) person
 - d) None of the given
35. Which statement is true for concrete class?
- a) It cannot be inherit from an abstract class
 - b) **It can be instantiated**
 - c) It cannot be instantiated
 - d) None of given
36. In c++, compiler can generate which of the following operators 'code,
- a) **==**
 - b) =
 - c) &
 - d) &&
37. In specialization we can,
- a) Replace child class with its base class
 - b) **Replace base class with its child class**
 - c) Replace both child and base classes interchangeably
 - d) None of the given option
38. In private inheritance derived class pointer can be assigned to base class pointer in,
- a) **Main function**
 - b) In derived class member and friend class function
 - c) In base class member and friend functions
 - d) None of the given option
39. When we access private members of the class from outside the class using ____ operator with its object pointer
- a. **->**
 - b. .
 - c. &
 - d. None of above
40. Which of the following access specifier ensure that base class member is accessible in derived class and Not outside of this class
- a. Public

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- b. Private
 - c. **Protected**
 - d. All of above
41. Static casting is
- a. **C++ way of calling base class function from derived class**
 - b. C way of calling base class function from derived class
 - c. Both of these
 - d. None of these
42. Suppose both derive and base classes have compilers generated copy constructors then derived class copy construct is
- a. Compiler generated default constructor
 - b. **Compiler generated copy constructor**
 - c. Both of these options are correct
 - d. None of these operators are correct
43. In case of protected inheritance, public members of the base class will be _____ in derived class
- a. Private
 - b. Public
 - c. **Protected**
 - d. Hidden
44. Methodologies to the development of reusable software related to _____
- a. Structured programming
 - b. Procedural programming
 - c. **Generic programming**
 - d. None of the given
45. A child class can call constructor of the parent call through
- a. **Its constructor initialization list**
 - b. Its constructor body
 - c. Both from its constructor initialization list or body
 - d. Cannot call the constructor of its parent class
46. In case of dynamic memory allocation in or class we should use
- a. Default assignment operator
 - b. **User defined assignment operator**
 - c. Default copy constructor
 - d. None of these
47. Consider the code below.
- ```
Class class1 {
Protected:
Int l;
};
Class class2: private calls 1{
};
```
- a) Public

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- b) Protected
- c) Private
- d) None of the given options

48. Consider the following two lines of code written for class student

1. Student subj1;
2. Student subj2(subj1)

In line 2 which constructor of student class will be called?

- a) Default constructor of student class
- b) Copy constructor of student class
- c) Both default and copy constructor of student class
- d) No constructor will be called

49. Which of the following is /are casting operator(s) inc++?

- a. Static\_cast
- b. Const\_cast
- c. Dynamic\_cast
- d. All of the given options

50. In case of private inheritance, private members of base class will be \_\_\_\_\_ in derived class?

- a. Private
- b. Public
- c. Protected
- d. Hidden

51. Consider the code below

```
class class1{
```

```
Private:
```

```
Void func1();
```

```
};
```

```
Class class2 : public class1{
```

```
};
```

Function func1 of class 1 is \_\_\_\_\_ in class2.

- a. Public
- b. Protected
- c. Private
- d. None of the given options

52. We can call base class assignment operator in derived class user defined assignment operator

- a. Implicitly
- b. Explicitly
- c. Using both of these options
- d. Using none of these options

53. A function call is resolved at run time in



- a. Non virtual members function
- b. Virtual member function**
- c. Both none-virtual and virtual member function
- d. None of given

54. If the user does not specify the type of inheritance, then the default type of inheritance is \_\_\_\_\_

- a. Public inheritance
- b. Private inheritance**
- c. Protected inheritance
- d. None of given

55. Consider the code below

```
class class1 {
public:
Void func1();
};
Class class2: protected class1 {
};
```

Function func1 of class 1 is \_\_\_\_\_ in class2.

- a. Public
- b. Protected**
- c. Private
- d. None of the given options

56. Which of the following causes run time binding?

- ▶ Declaring object of abstract class
- ▶ Declaring pointer of abstract class
- ▶ **Declaring overridden methods as non-virtual** (Page 226)
- ▶ None of the given

57. Which of the following is the best approach if it is required to have more than one functions having exactly same functionality and implemented on different data types?

- ▶ **Templates** (Page 256)
- ▶ Overloading
- ▶ Data hiding
- ▶ Encapsulation

58. A copy constructor is invoked when

- ▶ a function do not returns by value.
- ▶ **an argument is passed by value.** (Page 78)
- ▶ a function returns by reference.
- ▶ an argument is passed by reference.

59. Like template functions, a class template may not handle all the types successfully.

- ▶ **True** (Page 258)

► False

60. A class template may inherit from another class template.

► True (Page 288)

► False

61. By default the vector data items are initialized to \_\_\_\_\_

► 0 google

► 0.0

► 1

► null

62. In Private-----only member functions and friend classes or functions of a derived class can convert pointer or reference of derived object to that of parent object

► specialization

► inheritance (Page 216)

► abstraction

► composition

63. Which of the following is/are advantage[s] of generic programming?

► Reusability

► Writability

► Maintainability

► All of given (Page 256)

64. Template functions use \_\_\_\_\_ than ordinary functions.

► Greater Memory

► Lesser Memory

► Equal Memory

► None of the given options

65. Non Template Friend functions of a class are friends of \_\_\_\_\_instance/s of that class.

► All google

► One specific

► All instances of one date type

► None of the given options

66. A pointer to a base class can point to objects of a derived class.

► True google

► False

67. A template argument is preceded by the keyword\_\_\_\_\_.

- ▶ vector
  - ▶ [class](#)
  - ▶ template
  - ▶ type\*
68. Which one of the following terms must relate to polymorphism?
- ▶ Static allocation
  - ▶ Static typing
  - ▶ [Dynamic binding](#) (Page 239)
  - ▶ Dynamic allocation
69. Multiple inheritance can be of type
- ▶ Public
  - ▶ Private
  - ▶ Protected
  - ▶ [All of the given](#)
70. Assume a class Derv that is privately derived from class Base. An object of class Derv located in main() can access
- ▶ [public members of Derv](#)
  - ▶ protected members of Derv.
  - ▶ private members of Derv.
  - ▶ protected members of Base.
71. A function call is resolved at run-time in \_\_\_\_\_
- ▶ non-virtual member function.
  - ▶ [virtual member function.](#) (Page 239)
  - ▶ Both non-virtual member and virtual member function.
  - ▶ None of given
72. Two important STL associative containers are \_\_\_\_\_ and \_\_\_\_\_.
- ▶ [set, map](#)
  - ▶ sequence, mapping
  - ▶ set, multiset
  - ▶ set, map
73. An abstract class is useful when,
- ▶ We do not derive any class from it.
  - ▶ There are multiple paths from one derived class to another.
  - ▶ [We do not want to instantiate its object](#)
  - ▶ You want to defer the declaration of the class.
74. Suppose you create an uninitialized vector as follows:

`vector<int> evec;`

After adding the statment,

`evec.push_back(21);`

what will happen?

- ▶ The following statement will add an element to the start (the back) of evec and will initialize it with the value 21.
- ▶ The following statement will add an element to the center of evec and will reinitialize it with the value

21.

- ▶ The following statement will delete an element to the end (the back) of evec and will reinitialize it with the value 21.
- ▶ The following statement will add an element to the end (the back) of evec and initialize it with the value 21.

75. Default constructor is such constructor which either has no -----or if it has some parameters these have ----

- values

- ▶ Parameter, temporary
- ▶ Null, Parameter
- ▶ Parameter, default (Page 75)
- ▶ non of the given

76. Classes like TwoDimensionalShape and ThreeDimensionalShape would normally be concrete, while classes like Sphere and Cube would normally be abstract.

- ▶ True
- ▶ False

77. In order to define a class template, the first line of definition must be:

- ▶ `template <typename T>` (Page 257)
- ▶ `typename <template T>`
- ▶ `Template Class <ClassName>`
- ▶ `Class <Template T>`

78. A function template can not be overloaded by another function template.

- ▶ True
- ▶ False

79. Identify the correct way of declaring an object of user defined template class A for char type members?

- ▶ `A< char > obj;`
- ▶ `<char>A obj;`
- ▶ `A obj<char>;`
- ▶ `Obj <char> A;`

80. The user must define the operation of the copy constructor.

- ▶ True
- ▶ False

81. The find() algorithm

- ▶ finds matching sequences of elements in two containers.
- ▶ finds a container that matches a specified container.
- ▶ takes iterators as its first two arguments. (Object-Oriented Programming in C++)
- ▶ takes container elements as its first two arguments.

82. Compiler performs \_\_\_\_\_ type checking to diagnose type errors,

- ▶ Static (Page 261)
- ▶ Dynamic
- ▶ Bound
- ▶ Unbound

83. Vectors contain contiguous elements stored as a[an]\_\_\_\_\_.

- ▶ variable
- ▶ array (Page 306)
- ▶ function
- ▶ datatype

84. In a de-queue, (chosed the best option)

- ▶ data can be quickly inserted or deleted at any arbitrary location.
- ▶ data can be inserted or deleted at any arbitrary location, but the process is relatively slow.
- ▶ data can not be quickly inserted or deleted at either end.
- ▶ data can be inserted or deleted at either end, but the process is relatively slow.

85. Algorithms can only be implemented using STL containers.

- ▶ True
- ▶ False(Object-Oriented Programming in C++)



86. What is a class?

- ▶ A class is a section of computer memory containing objects.
- ▶ A class is a section of the hard disk reserved for object oriented programs
- ▶ A class is the part of an object that contains the variables.
- ▶ A class is a description of a kind of object. [Click here for detail](#)

87. Inheritance is a way to

- ▶ organize data.
- ▶ pass arguments to objects of classes.
- ▶ add features to existing classes without rewriting them. (Page 27)
- ▶ improve data-hiding and encapsulation.

88. We can use "this" pointer in the constructor in the body and even in the initialization list of any class if we are careful,

- ▶ True
- ▶ False

89. \_\_\_\_\_ and \_\_\_\_\_ methods may not be declared abstract.

- ▶ private,static [Click here for detail](#)
- ▶ private,public
- ▶ static,public
- ▶ none of given

90. \_\_\_\_\_ members are somewhere between public and private members. They are used in inheritance

- ▶ protected
- ▶ public
- ▶ private
- ▶ global (Page 187)

91. Which of these are examples of error handling techniques?

- ▶ Abnormal Termination
- ▶ Graceful Termination

- ▶ Return the illegal
- ▶ all of the given (Page 329)

92----- follow try block to catch the object thrown

- ▶ catch block (Page 333)
  - ▶ throw block
  - ▶ main block
  - ▶ non of the given

93. Graphical representation of the classes and objects is called object model it shows -----

- ▶ Class Name only
- ▶ Class Name and attributes
- ▶ Relationships of the objects and classes
- ▶ all of the given

94. Destructor can be overloaded

- ▶ True
- ▶ False(Page 92)

95. Which of the following is the best approach to implement generic algorithms with minimum number of coding lines?

- ▶ **Templates** (Page 256)
- ▶ Overloading
- ▶ Overriding
- ▶ Friend function/class

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96. Each try block can have\_\_\_\_\_no. of catch blocks.

- ▶ 1
- ▶ 2
- ▶ 3
- ▶ As many as necessary.

```
97. class DocElement
{
public:
 virtual void Print() { cout << "Generic element"; }
};
class Heading : public DocElement
{
public:
 void Print() { cout << "Heading element"; }
};
class Paragraph : public DocElement
{
public:
 void Print() { cout << "Paragraph element"; }
};
void main()
{
 DocElement * p = new Paragraph();
```

```
p->Print();
}
```

98. When you run this program, it will print out a single line to the console output.

What will be in that line?

Select one correct answer from the following list:

- ▶ Generic element ▶
- ▶ Heading element
- ▶ Paragraph element
- ▶ Nothing will be printed.

99. Suppose we have two derived classes from a single class, can we write a method with same name in both these derived classes? Choose the best option.

- ▶ No
- ▶ [Only if the two classes have the same name](#) (Page 204)
- ▶ Only if the main program does not declare both kinds
- ▶ Yes

100. When a virtual function is called by referencing a specific object by name and using the dot member selection operator (e.g., squareObject.draw()), the reference is resolved at compile time.

- ▶ True
- ▶ [False](#)

101. Considering the resolution order in which compiler search for functions in a program; the first priority is given to;

- ▶ general template
- ▶ partial specialization
- ▶ complete specialization
- ▶ [ordinary function](#) (Page 287)

102. One purpose of an iterator in the STL is to connect algorithms and containers.

- ▶ [True](#)
- ▶ False

103. In \_\_\_\_\_, a base class can be replaced by its derived class,

- ▶ [Sub-typing](#) (Page 31)
- ▶ Super-typing
- ▶ Multiple-typing
- ▶ Restricted-typing

104. one this pointer does not point to current object of any class,

- ▶ True
- ▶ False

105. Which of the following operator(s) take(s) one or no argument if overloaded?

- ▶ ++(Page 162)
- ▶ -
- ▶ +
- ▶ All of the given

106. Which of the following operators can not be overloaded?

- ▶ Scope resolution operator ( :: ) (Page 141)
- ▶ Insertion operator ( << )
- ▶ Extraction operator ( >> )
- ▶ The relation operator ( > )

107. Which of these are examples of error handling techniques ?

- ▶ Abnormal Termination
- ▶ Graceful Termination
- ▶ Return the illegal
- ▶ all of the given (Page 329)

108. \_\_\_\_\_ “is a” relationship

- ▶ Inheritance (Page 25)

- ▶ Polymorphism
- ▶ abstraction
- ▶ encapsulation

109. Virtual functions allow you to

- ▶ create an array of type pointer-to-base class that can hold pointers to derived classes.
- ▶ create functions that can never be accessed.
- ▶ group objects of different classes so they can all be accessed by the same function code.
- ▶ use the same function call to execute member functions of objects from different classes

110. The copy() algorithm returns an iterator to

- ▶ the last element copied from.
- ▶ the last element copied to.
- ▶ the element one past the last element copied from.
- ▶ the element one past the last element copied to.

111. If you define a vector v with the default constructor, and define another vector w with a one-argument constructor to a size of 11, and insert 3 elements into each of these vectors with push\_back(), then the size() member function will return \_\_\_\_\_ for v and \_\_\_\_\_ for w.

- ▶ 11 for v and 3 for w.
- ▶ 0 for v and 0 for w.
- ▶ 0 for v and 3 for w.
- ▶ 3 for v and 11 for w.

112. Which is not the Advantage of inheritance?

- ▶ providing class growth through natural selection. (Object-Oriented Programming in C++)
- ▶ facilitating class libraries.
- ▶ avoiding the rewriting of code.
- ▶ providing a useful conceptual framework.

113. Which type of inheritance is being represented by the following statement, class X : public A, public B { ..... };

- ▶ Single inheritance
- ▶ Multiple inheritance (Page 41)



- ▶ Double inheritance
- ▶ None of the given options

114. When we write a class template the first line must be:

- ▶ `template < class class_name>`
- ▶ `template < class data_type>`
- ▶ `template < class T >` (Page 257)

115. Function templates should be used where code and behavior must be identical.

- ▶ True (Page 262)
- ▶ False

116. The specialization pattern `<T*>` after the name says that this specialization is to be used for every,

- ▶ data type
- ▶ meta type
- ▶ virtual type
- ▶ pointer type (Page 286)

117. A range is often supplied to an algorithm by two \_\_\_\_\_ values.

- ▶ italic
- ▶ iteration (Object-Oriented Programming in C++)
- ▶ iterator
- ▶ None of given

118. Which of the following is an integral part of an object?

- ▶ State
- ▶ Behavior
- ▶ Unique identity
- ▶ All of the given (Page 12)

119. Consider the following statement

Cupboard has books

What is the relationship between Cupboard and books?

- ▶ Composition
- ▶ Aggregation
- ▶ Inheritance
- ▶ None of the given options

120. Which sentence clearly defines an object?

- ▶ one instance of a class. (Page 23)

- ▶ another word for a class.
- ▶ a class with static methods.
- ▶ a method that accesses class attributes.

121. \_\_\_\_\_, which means if A declares B as its friend it does NOT mean that A can access private data of B. It only means that B can access all data of A.

- ▶ Friendship is one way only
- ▶ Friendship is two way only
- ▶ NO Friendship between classes
- ▶ Any kind of friendship

122. The statement `objA=objB;` will cause a compiler error if the objects are of different classes.

- ▶ True
- ▶ False

123. Consider the call given below of an overloaded operator "+", *Rational\_number\_1* + *Rational\_number\_2*

Where *Rational\_number\_1* and *Rational\_number\_2* are the two objects of *Rational\_number* class (a user defined class). Identify which of the above two objects will be passed as an argument to the overloaded operator function?

- ▶ *Rational\_number\_1*
- ▶ *Rational\_number\_2*
- ▶ Both *Rational\_number\_1* & *Rational\_number\_2*
- ▶ any of the two objects, randomly

124. If a class D has been derived using protected inheritance from class B (If B is a protected base and D is derived class) then public and protected members of B -----accessed by member functions and friends of class D and classes derived from D

- ▶ can be
- ▶ cannot be
- ▶ does restrict to be
- ▶ not given

The type that is used to declare a reference or pointer is called its\_\_\_\_\_

- ▶ abstract type
- ▶ reference type
- ▶ [static type](#)
- ▶ defaulttype

125. How the information hidden within an object can be accessed?

- ▶ Through its interface
- ▶ Through its private data members
- ▶ [Through its private member functions](#)
- ▶ Through both public and private members

126. The sub-object's life is not dependant on the life of master class in\_\_\_\_\_.

- ▶ Separation
- ▶ Composition
- ▶ None of the given

127. Encapsulation means

Select correct option:

- ▶ Extending the behaviour of class in another class
- ▶ [Data and behaviour are tightly coupled within an entity](#) (Page 16)
- ▶ One entity takes all the attributes and operations of the other
- ▶ Taking out the common features and put those in a separate class

128. An STL container can not be used to,

- ▶ hold objects of class employee.
- ▶ store elements in a way that makes them quickly accessible.
- ▶ [compile c++ programs. \(Object-Oriented Programming in C++\)](#)
- ▶ organize the way objects are stored in memory

129. Which of the following may not be an integral part of an object?

- ▶ State
- ▶ Behavior
- ▶ [Protected data members \(Page 12\)](#)
- ▶ All of given

130. Public methods of base class can-----be accessed in its derived class

- ▶ [directly \(Page 179\) rep](#)
- ▶ inderectly
- ▶ simultaneously
- ▶ non of the given

131. What is true about function templates?

- ▶ The compiler generates only one copy of the function template
- ▶ The compiler generates a copy of function respective to each type of data (Page 256)
- ▶ The compiler can only generate copy for the int type data
- ▶ None of the given.

132. When the base class and the derived class have a member function with the same name, you must be more specific which function you want to call (using \_\_\_\_\_).

- ▶ scope resolution operator
- ▶ dot operator
- ▶ null operator
- ▶ Operator overloading

133.A template provides a convenient way to make a family of

- ▶ variables and data members
- ▶ functions and classes(Object-Oriented Programming in C++)
- ▶ classes and exceptions
- ▶ programs and algorithms

134.A non-virtual member function is defined in a base class and overridden in a derived class; if that function is called through a base-class pointer to a derived class object, the derived-class version is used.

- ▶ True
- ▶ False

135.If there is a pointer p to objects of a base class, and it contains the address of an object of a derived class, and both classes contain a nonvirtual member function, ding(), then the statement p->ding(); will cause the version of ding() in the \_\_\_\_\_ class to be executed.

- ▶ Base (Object-Oriented Programming in C++)
- ▶ Derived
- ▶ Abstract
- ▶ virtual

136. In case of multiple inheritance a derived class inherits,

- ▶ Only the public member functions of its base classes
- ▶ Only the public data members of its base classes
- ▶ Both public data members and member functions of all its base classes
- ▶ Data members and member functions of any two base classes

137. Which of the following is incorrect line regarding function template?

- ▶ `template<class T>`
- ▶ `template <typename U>`
- ▶ `Class<template T>` (Page 257)
- ▶ `template < class T, class U>`

138. Consider a class named Vehicle, which of the following can be the instance of class Vehicle?

- i. Car
  - ii. Computer
  - iii. Desk
  - iv. Ahmed
  - v. Bicycle
  - vi. Truck
- 1,4,5
  - 2,5,6
  - 1,2,3,6
  - 1, 5, 6 Conceptual

139. Consider the code below,

```
class Fred {
public:
Fred();
...
};
int main()
{
Fred a[10];
Fred* p = new Fred[10];
...
}
```

Select the best option,

- ▶ `Fred a[10];` calls the default constructor 09 times `Fred* p = new Fred[10];` calls the default constructor 10 times
- ▶ Produce an error

► Fred a[10]; calls the default constructor 11 times  
Fred\* p = new Fred[10]; calls the default constructor 11 times

► Fred a[10]; calls the default constructor 10 times  
Fred\* p = new Fred[10]; calls the default constructor 10 times

140. When a variable is define as static in a class then all object of this class,

- Have different copies of this variable
- Have same copy of this variable (Page 110)
- Can not access this variable
- None of given

141. Which of the following operators always takes no argument if overloaded?

- /
- -
- +
- ++ (Page 162)

142. It is sometimes useful to specify a class from which no objects will ever be created.

- True(Object-Oriented Programming in C++)
- False

143. When a subclass specifies an alternative definition for an attribute or method of its superclass, it is \_\_\_\_\_ the definition in the superclass.

- overload
- overriding (Page 34)
- copy riding
- none of given

144. The mechanism of selecting function at run time according to the nature of calling object is called,

- late binding
- static binding
- virtual binding
- None of the given options (Page 227)

145. Dynamic binding means that target function for a call is selected at run time

Which one of the following functions returns the total number of elements in a vector.

- length();
- size(); (Page 318)
- ele();
- veclen();



146. The main function of scope resolution operator (::) is,

- ▶ To define an object
- ▶ To define a data member
- ▶ [To link the definition of an identifier to its declaration](#)
- ▶ To make a class private

Google

147. When is a constructor called?

- ▶ Each time the constructor identifier is used in a program statement
- ▶ [During the instantiation of a new object \(Object-Oriented Programming in C++\)](#)
- ▶ During the construction of a new class
- ▶ At the beginning of any program execution

148. Associativity can be changed in operator overloading.

- ▶ True
- ▶ [False \(Page 141\)](#)

149. A normal C++ operator that acts in special ways on newly defined data types is said to be

- ▶ glorified.
- ▶ encapsulated.
- ▶ classified.
- ▶ [overloaded.](#)

150. Which operator can not be overloaded?

- ▶ The relation operator ( >= )
- ▶ Assignment operator ( = )
- ▶ Script operator ( [] )
- ▶ [Conditional operator \( ? : \) \(Page 141\)](#)

151. Suppose obj1 and obj2 are two objects of a user defined class A. An + operator is overloaded to add obj1 and obj2 using the function call obj1+obj2.

Identify the correct function prototype against the given call?

- ▶ A operator + ( A &obj);
- ▶ [int + operator\(\); \(Page 143\)](#)
- ▶ int operator (plus) ();
- ▶ A operator(A &obj3);

152. Suppose that the Test class does not have an overloaded assignment operator. What happens when an assignment  $a=b$ ; is given for two Test objects a and b?

- ▶ The automatic assignment operator is used
- ▶ **The copy constructor is used**
- ▶ Compiler error
- ▶ Run-time error

153. A static member function can be called, even when a class is not \_\_\_\_\_.

- ▶ Declared
- ▶ Define
- ▶ **Instantiated**
- ▶ Called

154. Identify which of the following overloaded operator function's declaration is appropriate for the given call?

**Rational\_number\_1 + 2.325**

Where Rational\_number\_1 is an object of user defined class Rational\_number.

- ▶ Rational\_number operator+( Rational\_number & obj);
- ▶ Rational\_number operator+(double& obj);
- ▶ Rational\_number operator+(Rational\_number &obj, double& num);
- ▶ operator+(double& obj);

155. What problem(s) may occur when we copy objects without using deep copy constructor?

- ▶ Dangling pointer
- ▶ Memory Leakage
- ▶ **All of the given**
- ▶ System crash

156. \_\_\_\_\_ provide the facility to access the data member.

- ▶ **accesser function**
- ▶ private function
- ▶ inline function
- ▶ None of the given

157. Constant objects cannot change their state,

- ▶ **True**
- ▶ False

158. The \_\_\_\_\_ relationship indicates that an object contains other objects.

- ▶ None of given
- ▶ **'has-a'**
- ▶ 'is-a'
- ▶ 'be-

159. Which one of the following features of OOP is used to derive a class from another?

- ▶ Encapsulation
- ▶ Polymorphism
- ▶ Data hiding
- ▶ **Inheritance**

160. \_\_\_\_\_ satisfy the condition of polymorphism

- ▶ Carbon
- ▶ Diamond
- ▶ Coal
- ▶ **all of the given**

161. A generalization-specialization relation between classes are implemented using

- ▶ data hiding
- ▶ friend classes
- ▶ encapsulation
- ▶ **inheritance**

162. The >= operator can be overloaded.

- ▶ **True**
- ▶ False

163. In order to free the memory occupied by the object, we use -----

- ▶ Constructor
- ▶ **Destructor**
- ▶ Shallow Copy
- ▶ Deep Copy

164. Which of the following is not an example of multiple inheritances?

- ▶ Mermaid
- ▶ **Woman**
- ▶ None of the given
- ▶ Amphibious Vehicle

165. Static variable can be initialized more than once.

- ▶ True
- ▶ **False**

166. A generic class showing all the common attributes and a behavior of other classes represents a very important feature in oop called -----

- ▶ **Inheritance**
- ▶ Encapsulation
- ▶ Polymorphism
- ▶ Abstraction

167. The parameters given in template definition other than those used for mentioning templates types are called \_\_\_\_

- a. Type Parameters
- b. **Non Type parameters**
- c. Default Type Parameters
- d. None of the given option

168. Which of the following is the correct way to define a template class X?

- e. `Class < typename T> class X {}`
- f. `Template class C {}`
- g. **`Template < typename T> class X {}`**
- h. `Typename < class T > class X {}`

169. In resolution order of function template, compiler searches for \_\_\_\_ in the end

- a. Complete specialization
- b. Partial specialization
- c. Ordinary function
- d. **Generic template**

170. \_\_\_\_ may inherit from a complete specialization

- a. Partial specialization
- b. Complete specialization
- c. Ordinary class
- d. **All of given options**

171. Which of the following may inherit from the ordinary class?

- a. Class template

- b. Partial specialization
- c. Complete specialization
- d. All of given option

172. Which of the following is a sequence container?

- a. Map
- b. Vector
- c. Set
- d. multiset

173. Each \_\_\_\_\_ of a template class by default becomes function template

- a. Data member
- b. Member function
- c. Type parameter
- d. Object

174. When we specialize a function template, it is called \_\_\_\_\_

- a. Function overloading
- b. Function overriding
- c. Function template overloading
- d. Function template overriding

175. Which of the following is NOT an associative container?

- a. Set
- b. Multiset
- c. Map
- d. Group

176A A \_\_\_\_\_ is a pointer which is declared outside the \_\_\_\_\_

- a. Container, cursor
- b. Cursor, container
- c. Container, class
- d. This, cursor

177. Which of the following are collectively referred to as the first class containers?

- a. Associative containers and containers adapters
- b. Sequence containers and associative containers
- c. Sequence containers and container adapters
- d. None of the given options

178. \_\_\_\_\_ class is a single class that provides functionality to operate on different type of data

- a. Friend
- b. Template
- c. Ordinary
- d. None of the give options

179. Which of the following can be passed as type argument to template?

- a. Primitive type
- b. User defined types

- c. Both primitive type and user defined types  
d. None of the given options
180. Which of the following is known as Dereference operator in C++?  
a. +b.  
\*  
c. &  
d. ::
181. We can change behavior of template using \_\_\_\_\_  
a. Template parameters  
b. Function parameters  
c. Class Templates  
d. none of the given options
182. Which of the following represents partial specialization?  
a. Template < int, char >  
b. Template < class T, Type t>  
c. Template<class T, Class U, int>  
d. Template< typename T, class W>
183. When we declare template function as friend of any class, then which of the followingspecialization(s) of template function also become friend of the class granting friendship?  
a. Explicit  
b. Implicit  
c. Partial  
d. All of the given options
184. In statement “template <class T, class U, int I = 5>” then non-type parameters is  
a. Class T  
b. Class U  
c. int I  
d. All of the given option
185. Which of the following is the correct syntax for passing two type argument to atemplate?  
a. template < type T, type U>  
b. template < typename T, typename U>  
c. template <type T,U>  
d. template Typename < T,U >
186. Which of the following represents complete specialization?  
a. Template < int, char >  
b. Template < class T, float>  
c. Template<class T, Class U , int>  
d. Template< class , class W>
187. Which of the following is/are the key component of standard template library?  
a. Containers  
b. Iterators



- c. Algorithm  
d. All of the given option
188. Which of the following statement is true about partial specialization?  
a. Function templates cannot have partial specialization  
b. Class template cannot have partial specialization  
c. Both class templates and function template can have partial specialization.  
d. None of the given
189. Which of the following is correct code portion to initialize static data member "value" of int type for class X?  
a. Template <class T> X::value =0;  
b. Template<class T>int X::value =0;  
c. Template<class T> int X < T >::value = 0;  
d. Template<class T> X < T >::value =0;
190. When we want to have exactly identical operations on different data type, \_\_\_\_\_ are used  
a. Function Template  
b. Function Overloading  
c. Function Overriding  
d. None of the given option
191. In order to define a class template the first line of definition must be:  
a. Template <typename T>  
b. typename < template T>  
c. Template class <ClassName>  
d. Class <Template T>
192. In case of template specialization, if compiler cannot find required complete specialization then it searches for some \_  
a. General template  
b. Complete template  
c. Partial specialization  
d. None of the given
193. Which of the following is the correct code to instantiate the object of given template vector class of int type?  
Template <class T>  
class vector  
{  
};  
a. Vector obj <int>;  
b. Vector obj <> int;  
c. Vector <int>obj;  
d. Vector int obj;
194. Which of the following types of container is NOT provided by standard Template Library?  
a. Sequence containers

b. **Structured containers**

c. Associative containers

d. Container adapters

195. A class template \_\_\_\_\_

a. **Facilitates reuse of class**

b. Does not facilitate reuse of class

c. Does not support generic methods

d. Does not support static members

196. Suppose a template class "test" has a static data member, how many copies of this static data member will be created when template class is instantiated by writing following line of C++ code?

Test <char>x,y,z

a. Zero

b. One **Correct**

c. Two

197. STL stands for:

a. Stack Template Library

b. **Standard Template Library**

c. Structured Template Library

d. Secure Template Library

198. Which of the following is not a sequence container?

a. Vector

b. Deque **c. Map**

d. List

199. Suppose a template class "test" has a static data member, how many copies of this static data member will be created when template class is instantiated by writing following line of C++ code?

Test <int>x; Test <char>y;

a. Zero

b. One **c. Two**

d. Three

200. Suppose we have a template class "Test" which has a static data member "data" which is initialized with 0. What will be output when following code is written in main() function for given class?

Test<int> a, b; a.data = 5;

b.data = 7; cout<<a.data;

a. 0

b. ~~1e-5~~

**d. 7 Correct**

201. In resolution order, highest priority is given to \_\_\_\_\_ in template specialization.

a. General template

b. **Complete specialization**

c. Partial specialization

d. None of the given options

202. Which of the following is a constrained version of some first-class container?

- a. Sequence containers
- b. Associative containers
- c. Container Adapter
- d. None of the given options

Question No: 1 ( Marks: 1 ) - Please choose one

Which of the following causes run time binding?

- ▶ Declaring object of abstract class
- ▶ Declaring pointer of abstract class
- ▶ Declaring overridden methods as non-virtual (Page 226)
- ▶ None of the given

Question No: 2 ( Marks: 1 ) - Please choose one Which of the following is the best approach if it is required to have more than one functions having exactly same functionality and implemented on different data types?

▶ Templates (Page 256)

- ▶ Overloading
- ▶ Data hiding
- ▶ Encapsulation

Question No: 3 ( Marks: 1 ) - Please choose one A copy constructor is invoked when

- ▶ a function do not returns by value.
- ▶ an argument is passed by value. (Page 78)
- ▶ a function returns by reference.
- ▶ an argument is passed by reference.

Question No: 4 ( Marks: 1 ) - Please choose one Like template functions, a class template may not handle all the types successfully

▶ True (Page 258)

- ▶ False 2

Question No: 6 ( Marks: 1 ) - Please choose one A class template may inherit from another class template

▶ True (Page 288)

- ▶ False

Question No: 7 ( Marks: 1 ) - Please choose one By default the vector data items are initialized to

▶ 0 Click here for detail

- ▶ 0.0
- ▶ 1
- ▶ null

Question No: 8 ( Marks: 1 ) - Please choose one In Private -----only member functions and friend classes or functions of a derived class can convert pointer or reference of derived object to that of parent object

- ▶ specialization
- ▶ inheritance (Page 216)
- ▶ abstraction
- ▶ composition Question

No: 9 ( Marks: 1 ) - Please choose one Which of the following is/are advantage[s] of generic programming?

- ▶ Reusability
- ▶ Writability
- ▶ Maintainability
- ▶ All of given (Page 256)

Question No: 10 ( Marks: 1 ) - Please choose one Template functions use \_\_\_\_\_ than ordinary functions

- ▶ Greater Memory
- ▶ Lesser Memory
- ▶ Equal Memory
- ▶ None of the given options

Question No: 11 ( Marks: 1 ) - Please choose one Non Template Friend functions of a class are friends of \_\_\_\_\_instance/s of that class.

- ▶ All Click here for detail
- ▶ One specific
- ▶ All instances of one date type
- ▶ None of the given options 3

Question No: 12 ( Marks: 1 ) - Please choose one A copy constructor is invoked when

- ▶ a function do not returns by value.
- ▶ an argument is passed by value. (Page 78)
- ▶ a function returns by reference.
- ▶ an argument is passed by reference

. Question No: 13 ( Marks: 1 ) - Please choose one A pointer to a base class can point to objects of a derived class.

- ▶ True Click here for detail
- ▶ False

Question No: 14 ( Marks: 1 ) - Please choose one A template argument is preceded by the keyword

- ▶ vector
- ▶ class Click here for Detail
- ▶ template
- ▶ type\*

Question No: 15 ( Marks: 1 ) - Please choose one Which one of the following terms must relate to polymorphism?

- ▶ Static allocation
- ▶ Static typing
- ▶ Dynamic binding (Page 239)
- ▶ Dynamic allocation

Question No: 16 ( Marks: 1 ) - Please choose one Multiple inheritance can be of type

- ▶ Public
- ▶ Private
- ▶ Protected
- ▶ All of the given Click here for detail

Question No: 17 ( Marks: 1 ) - Please choose one Assume a class Derv that is privately derived from class Base. An object of class Derv located in main() can access

- ▶ public members of Derv. (Object-Oriented Programming in C++)
- ▶ protected members of Derv.
- ▶ private members of Derv.
- ▶ protected members of Base. 4

Question No: 18 ( Marks: 1 ) - Please choose one A copy constructor is invoked when

- ▶ a function do not returns by value.
- ▶ an argument is passed by value. (Page 78) (rep)
- ▶ a function returns by reference.
- ▶ an argument is passed by reference.

Question No: 19 ( Marks: 1 ) - Please choose one A function call is resolved at run-time in \_\_\_\_\_

- ▶ non-virtual member function.
- ▶ virtual member function. (Page 239)
- ▶ Both non-virtual member and virtual member function.
- ▶ None of given

Question No: 20 ( Marks: 1 ) - Please choose one Two important STL associative containers are \_\_\_\_\_ and \_\_\_\_\_.

- ▶ set,map (Object-Oriented Programming in C++)
- ▶ sequence,mapping
- ▶ setmet,multipule
- ▶ sit,mat

Question No: 21 ( Marks: 1 ) - Please choose one An abstract class is useful when,

- ▶ We do not derive any class from it.
- ▶ There are multiple paths from one derived class to another.
- ▶ We do not want to instantiate its object. (Object-Oriented Programming in C++)
- ▶ You want to defer the declaration of the class.

Question No: 22 ( Marks: 1 ) - Please choose one Which of the following is/are advantage[s] of generic programming?

- ▶ Reusability
- ▶ Writability
- ▶ Maintainability
- ▶ All of given (Page 256) rep

Question No: 23 ( Marks: 1 ) - Please choose one By default the vector data items are initialized to \_\_\_\_\_

- ▶ 0 Click here for Detail rep
- ▶ 0.0
- ▶ 1
- ▶ null 5

Question No: 24 ( Marks: 1 ) - Please choose one Suppose you create an uninitialized vector as follows: vector evec; After adding the statment, evec.push\_back(21); what will happen?

- ▶ The following statement will add an element to the start (the back) of evec and will initialize it with the value 21.
- ▶ The following statement will add an element to the center of evec and will reinitialize it with the value 21.
- ▶ The following statement will delete an element to the end (the back) of evec and will reinitialize it with the value 21.
- ▶ The following statement will add an element to the end (the back) of evec and initialize it with the value 21. Click here for detail

Question No: 25 ( Marks: 1 ) - Please choose one Default constructor is such constructor which either has no -----or if it has some parameters these have----- values

- ▶ Parameter, temporary
- ▶ Null, Parameter
- ▶ Parameter, default (Page 75)
- ▶ non of the given

Question No: 26 ( Marks: 1 ) - Please choose one Which of the following is the best approach if it is required to have more than one functions having exactly same functionality and implemented on different data types?

- ▶ Templates (Page 256) rep
- ▶ Overloading
- ▶ Data hiding
- ▶ Encapsulation

Question No: 27 ( Marks: 1 ) Describe the way to declare a template class as a friend of any class.

Question No: 28 ( Marks: 1 ) Classes like TwoDimensionalShape and ThreeDimensionalShape would normally be concrete, while classes like Sphere and Cube would normally be abstract.

- ▶ True Click here for detail
- ▶ False 6

Question No: 29 ( Marks: 1 ) In order to define a class template, the first line of definition must be:

- ▶ template (Page 257)
- ▶ typename
- ▶ Template Class
- ▶ Class

Question No: 30 ( Marks: 1 ) In case of multiple inheritance a derived class inherits,

- ▶ Only the public member functions of its base classes
- ▶ Only the public data members of its base classes
- ▶ Both public data members and member functions of all its base classes Click here for detail
- ▶ Data members and member functions of any two base classes

Question No: 31 ( Marks: 1 ) In Private----- only member functions and friend classes or functions of a derived class can convert pointer or reference of derived object to that of parent object

- ▶ specialization
- ▶ inheritance (Page 216) rep



- ▶ abstraction
- ▶ composition

Question No: 32 ( Marks: 1 ) Which of the following is the best approach if it is required to have more than one functions having exactly same functionality and implemented on different data types?

- ▶ Templates (Page 256) rep
- ▶ Overloading
- ▶ Data hiding
- ▶ Encapsulation

FINALTERM EXAMINATION Spring 2010 CS304- Object Oriented Programming

Question No: 1 ( Marks: 1 ) - Please choose one A template argument is preceded by the keyword

- ▶ vector
- ▶ class (Object-Oriented Programming in C++)
- ▶ template
- ▶ type\*

Question No: 2 ( Marks: 1 ) - Please choose one Which of the following causes run time binding?

- ▶ Declaring object of abstract class
- ▶ Declaring pointer of abstract class
- ▶ Declaring overridden methods as non-virtual (Page 226)
- ▶ None of the given

Question No: 3 ( Marks: 1 ) - Please choose one A function template can not be overloaded by another function template.

- ▶ True (Object-Oriented Programming in C++)
- ▶ False

Question No: 4 ( Marks: 1 ) - Please choose one Which of the following is the best approach if it is required to have more than one functions having exactly same functionality and implemented on different data types?

- ▶ Templates (Page 256) rep
- ▶ Overloading
- ▶ Data hiding

▶ Encapsulation Question No: 5 ( Marks: 1 ) - Please choose one Identify the correct way of declaring an object of user defined template class A for char type members?

- ▶ A< char > obj; (Object-Oriented Programming in C++)
- ▶ A obj;
- ▶ A obj;
- ▶ Obj A;

Question No: 6 ( Marks: 1 ) - Please choose one The user must define the operation of the copy constructor.

- ▶ True Click here for detail
- ▶ False

Question No: 7 ( Marks: 1 ) - Please choose one Template functions use \_\_\_\_\_ than ordinary functions.

- ▶ Greater Memory
- ▶ Lesser Memory
- ▶ Equal Memory
- ▶ None of the given options

Question No: 8 ( Marks: 1 ) - Please choose one The find() algorithm

- ▶ finds matching sequences of elements in two containers.
- ▶ finds a container that matches a specified container.
- ▶ takes iterators as its first two arguments. (Object-Oriented Programming in C++)
- ▶ takes container elements as its first two arguments.

Question No: 9 ( Marks: 1 ) - Please choose one Compiler performs \_\_\_\_\_ type checking to diagnose type errors,

- ▶ Static (Page 261)
- ▶ Dynamic
- ▶ Bound
- ▶ Unbound

Question No: 10 ( Marks: 1 ) - Please choose one Which of the following is/are advantage[s] of generic programming?

- ▶ Reusability
- ▶ Writability
- ▶ Maintainability
- ▶ All of given (Page 256) rep

Question No: 11 ( Marks: 1 ) - Please choose one Vectors contain contiguous elements stored as a[an]\_\_\_\_\_.

- ▶ variable
- ▶ array (Page 306)
- ▶ function
- ▶ datatype

Question No: 12 ( Marks: 1 ) - Please choose one Suppose you create an uninitialized vector as follows: vector evec; After adding the statment, evec.push\_back(21); what will happen?

- ▶ The following statement will add an element to the start (the back) of evec and will initialize it with the value 21.
- ▶ The following statement will add an element to the center of evec and will reinitialize it with the value 21.
- ▶ The following statement will delete an element to the end (the back) of evec and will reinitialize it with the value 21.
- ▶ The following statement will add an element to the end (the back) of evec and initialize it with the value 21. Click here for detail

Question No: 13 ( Marks: 1 ) - Please choose one In a de-queue, (chose the best option)

- ▶ data can be quickly inserted or deleted at any arbitrary location.
- ▶ data can be inserted or deleted at any arbitrary location, but the process is relatively slow. (Object-Oriented Programming in C++)
- ▶ data can not be quickly inserted or deleted at either end.
- ▶ data can be inserted or deleted at either end, but the process is relatively slow.

Question No: 14 ( Marks: 1 ) - Please choose one Algorithms can only be implemented using STL containers.

- ▶ True
- ▶ False (Object-Oriented Programming in C++)

Question No: 15 ( Marks: 1 ) - Please choose one What is a class?

- ▶ A class is a section of computer memory containing objects.
- ▶ A class is a section of the hard disk reserved for object oriented programs
- ▶ A class is the part of an object that contains the variables.

► A class is a description of a kind of object. Click here for detail

Question No: 16 ( Marks: 1 ) - Please choose one Inheritance is a way to

- organize data.
- pass arguments to objects of classes.
- add features to existing classes without rewriting them. (Page 27)
- improve data-hiding and encapsulation.

Question No: 17 ( Marks: 1 ) - Please choose one We can use "this" pointer in the constructor in the body and even in the initialization list of any class if we are careful,

- True Click here for detail
- False

Question No: 18 ( Marks: 1 ) - Please choose one \_\_\_\_\_ and \_\_\_\_\_ methods may not be declared abstract.

- private,static Click here for detail
- private,public
- static,public
- none of given

Question No: 19 ( Marks: 1 ) - Please choose one Default constructor is such constructor which either has no -----or if it has some parameters these have----- values

- Parameter, temporary
- Null, Parameter
- Parameter, default (Page 75) rep
- non of the given

Question No: 20 ( Marks: 1 ) - Please choose one Public methods of base class can----- be accessed in its derived class

- directly (Page 179)
- inderectly
- simultaneously
- non of the given

Question No: 21 ( Marks: 1 ) - Please choose one The type that is used to declare a reference or pointer is called its -----

- default type
- static type (Page 185)
- abstract type
- reference type

Question No: 22 ( Marks: 1 ) - Please choose one ----- members are somewhere between public and private members. They are used in inheritance

- protected (Page 187)
- public
- private
- global

Question No: 23 ( Marks: 1 ) - Please choose one Which of these are examples of error handling techniques?

- Abnormal Termination
- Graceful Termination
- Return the illegal
- all of the given (Page 329)

Question No: 24 ( Marks: 1 ) - Please choose one ----- follow try block to catch the object thrown

- ▶ catch block (Page 333)
- ▶ throw block
- ▶ main block
- ▶ non of the given

Question No: 25 ( Marks: 1 ) - Please choose one Graphical representation of the classes and objects is called object model it shows -----

- ▶ Class Name only
- ▶ Class Name and attributes
- ▶ Relationships of the objects and classes
- ▶ all of the given Click here for detail

Question No: 26 ( Marks: 1 ) - Please choose one Destructor can be overloaded

- ▶ True
- ▶ False (Page 92)

FINALTERM EXAMINATION Spring 2010 CS304- Object Oriented Programming (Session - 3) 12

Question No: 1 ( Marks: 1 ) - Please choose one Which one of the following terms must relate to polymorphism?

- ▶ Static allocation
- ▶ Static typing
- ▶ Dynamic binding (Page 239)
- ▶ Dynamic allocation

Question No: 2 ( Marks: 1 ) - Please choose one Which of the following causes run time binding?

- ▶ Declaring object of abstract class
- ▶ Declaring pointer of abstract class
- ▶ Declaring overridden methods as non-virtual (Page 226)
- ▶ None of the given

Question No: 3 ( Marks: 1 ) - Please choose one Which of the following is the best approach if it is required to have more than one functions having exactly same functionality and implemented on different data types?

- ▶ Templates (Page 256) rep
- ▶ Overloading
- ▶ Data hiding
- ▶ Encapsulation

Question No: 4 ( Marks: 1 ) - Please choose one Which of the following is the best approach to implement generic algorithms with minimum number of coding lines?

- ▶ Templates (Page 256)
- ▶ Overloading
- ▶ Overriding
- ▶ Friend function/class

Question No: 5 ( Marks: 1 ) - Please choose one Like template functions, a class template may not handle all the types successfully.

- ▶ True (Page 258) rep
- ▶ False

Question No: 6 ( Marks: 1 ) - Please choose one A class template may inherit from another class template.

- ▶ True (Page 288) rep

► False 13

Question No: 7 ( Marks: 1 ) - Please choose one Assume a class Derv that is privately derived from class Base. An object of class Derv located in main() can access

- public members of Derv. (Object-Oriented Programming in C++) rep
- protected members of Derv.
- private members of Derv.
- protected members of Base

. Question No: 8 ( Marks: 1 ) - Please choose one A copy constructor is invoked when

- a function do not returns by value.
- an argument is passed by value. (Page 78) rep
- a function returns by reference.
- an argument is passed by reference.

Question No: 9 ( Marks: 1 ) - Please choose one Each try block can have \_\_\_\_\_no. of catch blocks.

- 1
- 2
- 3

► As many as necessary. Click here for detail A single try block can have multiple catch blocks but only one finally block.

Question No: 10 ( Marks: 1 ) - Please choose one class DocElement { public: virtual void Print() { cout << "Generic element"; } }; class Heading : public DocElement { public: void Print() { cout << "Heading element"; } }; class Paragraph : public DocElement { public: void Print() { cout << "Paragraph element"; } }; void main() { DocElement \* p = new Paragraph(); 14 p->Print(); } When you run this program, it will print out a single line to the console output. What will be in that line? Select one correct answer from the following list:

- Generic element
- Heading element
- Paragraph element
- Nothing will be printed.

Question No: 11 ( Marks: 1 ) - Please choose one Suppose we have two derived classes from a single class, can we write a method with same name in both these derived classes? Choose the best option.

- No
- Only if the two classes have the same name (Page 204)
- Only if the main program does not declare both kinds
- Yes

Question No: 12 ( Marks: 1 ) - Please choose one When a virtual function is called by referencing a specific object by name and using the dot member selection operator (e.g., squareObject.draw()), the reference is resolved at compile time.

- True
- False (Object-Oriented Programming in C++) (the function invocation is resolved at compile time)

Question No: 13 ( Marks: 1 ) - Please choose one Considering the resolution order in which compiler search for functions in a program; the first priority is given to; the first priority is given to,

- general template
- partial specialization
- complete specialization

► ordinary function (Page 287)

Question No: 14 ( Marks: 1 ) - Please choose one Vectors contain contiguous elements stored as a[an]\_\_\_\_\_.

► variable

► array (Page 306) rep

► function

► datatype

Question No: 15 ( Marks: 1 ) - Please choose one By default the vector data items are initialized to

► 0 Click here for detail rep

► 0.0

► 1

► null

Question No: 16 ( Marks: 1 ) - Please choose one One purpose of an iterator in the STL is to connect algorithms and containers.

► True (Object-Oriented Programming in C++)

► False

Question No: 17 ( Marks: 1 ) - Please choose one Algorithms can only be implemented using STL containers.

► True

► False (Object-Oriented Programming in C++) rep

Question No: 18 ( Marks: 1 ) - Please choose one In \_\_\_\_\_, a base class can be replaced by its derived class,

► Sub-typing (Page 31)

► Super-typing

► Multiple-typing

► Restricted-typing

Question No: 19 ( Marks: 1 ) - Please choose one this pointer does not point to current object of any class,

► True

► False Click here for detail The this pointer is a hidden pointer inside every class member function that points to the class object the member function is working with.

Question No: 20 ( Marks: 1 ) - Please choose one Which of the following operator(s) take(s) one or no argument if overloaded?

► ++ (Page 162)

► -

► +

► All of the given

16 Question No: 21 ( Marks: 1 ) - Please choose one Which of the following operators can not be overloaded?

► Scope resolution operator ( :: ) (Page 141)

► Insertion operator ( << )

► Extraction operator ( >> )

► The relation operator ( > )

Question No: 22 ( Marks: 1 ) - Please choose one The type that is used to declare a reference or pointer is called its -----

► default type



- ▶ static type (Page 185) rep
- ▶ abstract type
- ▶ reference type

Question No: 23 ( Marks: 1 ) - Please choose one ----- members are somewhere between public and private members. They are used in inheritance

- ▶ protected (Page 187) rep
- ▶ public
- ▶ private
- ▶ global

Question No: 24 ( Marks: 1 ) - Please choose one Which of these are examples of error handling techniques ?

- ▶ Abnormal Termination
- ▶ Graceful Termination
- ▶ Return the illegal
- ▶ all of the given (Page 329)

Question No: 25 ( Marks: 1 ) - Please choose one \_\_\_\_\_ "is a" relationship

- ▶ Inheritance (Page 25)
- ▶ Polymorphism
- ▶ abstraction ▶
- encapsulation

Question No: 26 ( Marks: 1 ) - Please choose one Graphical representation of the classes and objects is called object model it shows -----

- ▶ Class Name only
- ▶ Class Name and attributes
- ▶ Relationships of the objects and classes
- ▶ all of the given Click here for detail rep

17 FINALTERM EXAMINATION Spring 2010 CS304- Object Oriented Programming (Session - 4)

Question No: 1 ( Marks: 1 ) - Please choose one Classes like TwoDimensionalShape and ThreeDimensionalShape would normally be concrete, while classes like Sphere and Cube would normally be abstract.

- ▶ True Click here for detail rep
- ▶ False

Question No: 2 ( Marks: 1 ) - Please choose one Virtual functions allow you to

- ▶ create an array of type pointer-to-base class that can hold pointers to derived classes.
- ▶ create functions that can never be accessed.
- ▶ group objects of different classes so they can all be accessed by the same function code.
- ▶ use the same function call to execute member functions of objects from different classes

(Object-Oriented Programming in C++)

Question No: 3 ( Marks: 1 ) - Please choose one .

- ▶ True Click here for detail rep
- ▶ False

Question No: 4 ( Marks: 1 ) - Please choose one A copy constructor is invoked when

- ▶ a function do not returns by value.
- ▶ an argument is passed by value. (Page 78) rep
- ▶ a function returns by reference.
- ▶ an argument is passed by reference.

Question No: 5 ( Marks: 1 ) - Please choose one Each try block can have \_\_\_\_\_no. of catch blocks.

- ▶ 1
- ▶ 2
- ▶ 3
- ▶ As many as necessary. Click here for detail rep

18 Question No: 6 ( Marks: 1 ) - Please choose one Non Template Friend functions of a class are friends of \_\_\_\_\_instance/s of

- ▶ All Click here for detail rep
- ▶ One specific
- ▶ All instances of one date type
- ▶ None of the given options

Question No: 7 ( Marks: 1 ) - Please choose one Template functions use \_\_\_\_\_than ordinary functions.

- ▶ Greater Memory
- ▶ Lesser Memory
- ▶ Equal Memory
- ▶ None of the given options

Question No: 8 ( Marks: 1 ) - Please choose one The find() algorithm

- ▶ finds matching sequences of elements in two containers.
- ▶ finds a container that matches a specified container.
- ▶ takes iterators as its first two arguments. (Object-Oriented Programming in C++)
- ▶ takes container elements as its first two arguments.

Question No: 9 ( Marks: 1 ) - Please choose one The copy() algorithm returns an iterator to

- ▶ the last element copied from.
- ▶ the last element copied to.
- ▶ the element one past the last element copied from.
- ▶ the element one past the last element copied to. (Object-Oriented Programming in C++)

Question No: 10 ( Marks: 1 ) - Please choose one If you define a vector v with the default constructor, and define another vector w with a one-argument constructor to a size of 11, and insert 3 elements into each of these vectors with push\_back(), then the size() member function will return \_\_\_\_\_ for v and \_\_\_\_\_for w.

- ▶ 11 for v and 3 for w.
- ▶ 0 for v and 0 for w.
- ▶ 0 for v and 3 for w.
- ▶ 3 for v and 11 for w. (Object-Oriented Programming in C++)

19 Question No: 11 ( Marks: 1 ) - Please choose one Which is not the Advantage of inheritance?

- ▶ providing class growth through natural selection. (Object-Oriented Programming in C++)
- ▶ facilitating class libraries.
- ▶ avoiding the rewriting of code.
- ▶ providing a useful conceptual framework.

Question No: 12 ( Marks: 1 ) - Please choose one class DocElement { public: virtual void Print() { cout << "Generic element"; } }; class Heading : public DocElement { public: void Print() { cout << "Heading element"; } }; class Paragraph : public DocElement { public: void Print() { cout << "Paragraph element"; } }; void main() { DocElement \* p = new Paragraph(); p->Print(); } When you run this program, it will print out a single line to the console output. What will be in that line? Select one correct answer from the following list:

- ▶ Generic element
- ▶ Heading element
- ▶ Paragraph element

▶ Nothing will be printed. 20

Question No: 13 ( Marks: 1 ) - Please choose one Which type of inheritance is being represented by the following statement, class X : public A, public B { .....};

- ▶ Single inheritance
- ▶ Multiple inheritance (Page 41)
- ▶ Double inheritance
- ▶ None of the given options

Question No: 14 ( Marks: 1 ) - Please choose one When we write a class template the first line must be:

- ▶ template < class class\_name>
- ▶ template < class data\_type>
- ▶ template < class T > (Page 257) Here T can be replaced with any name but it is preferable.
- ▶ class class-name() class template

Question No: 15 ( Marks: 1 ) - Please choose one Function templates should be used where code and behavior must be identical.

- ▶ True (Page 262)
- ▶ False

Question No: 16 ( Marks: 1 ) - Please choose one Which of the following is/are advantage[s] of generic programming?

- ▶ Reusability
- ▶ Writability
- ▶ Maintainability
- ▶ All of given (Page 256) rep

Question No: 17 ( Marks: 1 ) - Please choose one The specialization pattern after the name says that this specialization is to be used for every,

- ▶ data type
- ▶ meta type
- ▶ virtual type
- ▶ pointer type (Page 286)

Question No: 18 ( Marks: 1 ) - Please choose one A range is often supplied to an algorithm by two \_\_\_\_\_ values.

- ▶ italic
- ▶ iteration (Object-Oriented Programming in C++)
- ▶ iterator
- ▶ None of given 21

Question No: 19 ( Marks: 1 ) - Please choose one Which of the following is an integral part of an object?

- ▶ State
- ▶ Behavior
- ▶ Unique identity
- ▶ All of the given (Page 12)

Question No: 20 ( Marks: 1 ) - Please choose one Consider the following statement Cupboard has books What is the relationship between Cupboard and books?

- ▶ Composition

- ▶ Aggregation
- ▶ Inheritance
- ▶ None of the given options

Question No: 21 ( Marks: 1 ) - Please choose one Which sentence clearly defines an object?

- ▶ one instance of a class. (Page 23)
- ▶ another word for a class.
- ▶ a class with static methods.
- ▶ a method that accesses class attributes.

Question No: 22 ( Marks: 1 ) - Please choose one \_\_\_\_\_, which means if A declares B as its friend it does NOT mean that A can access private data of B. It only means that B can access all data of A.

- ▶ Friendship is one way only Click here for detail
- ▶ Friendship is two way only
- ▶ NO Friendship between classes
- ▶ Any kind of friendship

Question No: 23 ( Marks: 1 ) - Please choose one The statement objA=objB; will cause a compiler error if the objects are of different classes.

- ▶ True
- ▶ False (Object-Oriented Programming in C++) 22

Question No: 24 ( Marks: 1 ) - Please choose one Consider the call given below of an overloaded operator "+", Rational\_number\_1 + Rational\_number\_2 Where Rational\_number\_1 and Rational\_number\_2 are the two objects of Rational\_number class (a user defined class). Identify which of the above two objects will be passed as an argument to the overloaded operator function?

- ▶ Rational\_number\_1
- ▶ Rational\_number\_2
- ▶ Both Rational\_number\_1 & Rational\_number\_2
- ▶ any of the two objects, randomly

Question No: 25 ( Marks: 1 ) - Please choose one If a class D has been derived using protected inheritance from class B (If B is a protected base and D is derived class) then public and protected members of B----- accessed by member functions and friends of class D and classes derived from D

- ▶ can be Click here for detail
- ▶ cannot be
- ▶ does restrict to be
- ▶ not given

Question No: 26 ( Marks: 1 ) - Please choose one In Private -----only member functions and friend classes or functions of a derived class can convert pointer or reference of derived object to that of parent object

- ▶ specialization
- ▶ inheritance (Page 216) rep
- ▶ abstraction
- ▶ composition

FINAL TERM EXAMINATION SPRING 2010

Question No: 1 ( Marks: 1 ) - Please choose one Classes like TwoDimensionalShape and ThreeDimensionalShape would normally be concrete, while classes like Sphere and Cube would normally be abstract.

- ▶ True Click here for detail rep

► False 23

Question No: 2 ( Marks: 1 ) - Please choose one Each try block can have \_\_\_\_\_no. of catch blocks.

► 1

► 2

► 3

► As many as necessary. Click here detail

Question No: 3 ( Marks: 1 ) - Please choose one Function templates should be used where code and behavior must be identical.

► True (Page 262)

► False

Question No: 4 ( Marks: 1 ) - Please choose one Consider the following statement Cupboard has books What is the relationship between Cupboard and books?

► Composition

► Aggregation

► Inheritance

► None of the given options

Question No: 5 ( Marks: 1 ) - Please choose one Identify the correct way of declaring an object of user defined template class A for char type members?

► A< char > obj; (Object-Oriented Programming in C++)

► A obj;

► A obj;

► Obj A;

Question No: 6 ( Marks: 1 ) - Please choose one The user must define the operation of the copy constructor.

► True Click here for detail

► False

Question No: 7 ( Marks: 1 ) - Please choose one Default constructor is such constructor which either has no -----or if it has some parameters these have----- values

► Parameter, temporary

► Null, Parameter

► Parameter, default (Page 75) rep

► non of the given

Question No: 8 ( Marks: 1 ) - Please choose one The type that is used to declare a reference or pointer is called its -----

► default type

► static type (Page 185)

► abstract type

► reference type

24 Question No: 9 ( Marks: 1 ) - Please choose one How the information hidden within an object can be accessed?

► Through its interface

► Through its private data members

► Through its private member functions

► Through both public and private members

Question No: 10 ( Marks: 1 ) - Please choose one The sub-object's life is not dependant on the life of master class in\_\_\_\_\_.

- ▶ Separation
- ▶ Composition
- ▶ Aggregation (Page 134)
- ▶ None of the given

Question No: 11 ( Marks: 1 ) - Please choose one Encapsulation means Select correct option:

- ▶ Extending the behaviour of class in another class
- ▶ Data and behaviour are tightly coupled within an entity (Page 16)
  - ▶ One entity takes all the attributes and operations of the other
- ▶ Taking out the common features and put those in a separate class

Question No: 12 ( Marks: 1 ) - Please choose one Algorithms can only be implemented using STL containers.

- ▶ True
- ▶ False (Object-Oriented Programming in C++) rep

Question No: 13 ( Marks: 1 ) - Please choose one When we write a class template the first line must be:

- ▶ `template < class class_name>`
- ▶ `template < class data_type>`
- ▶ `template < class T >` (Page 257) rep Here T can be replaced with any name but it is preferable.
- ▶ `class class-name() class template`

Question No: 14 ( Marks: 1 ) - Please choose one An STL container can not be used to,

- ▶ hold objects of class employee.
- ▶ store elements in a way that makes them quickly accessible.
- ▶ compile c++ programs. (Object-Oriented Programming in C++)
- ▶ organize the way objects are stored in memory 25

Question No: 15 ( Marks: 1 ) - Please choose one \_\_\_\_\_, which means if A declares B as its friend it does NOT mean that A can access private data of B. It only means that B can access all data of A.

- ▶ Friendship is one way only Click here for detail
- ▶ Friendship is two way only
- ▶ NO Friendship between classes
- ▶ Any kind of friendship

Question No: 16 ( Marks: 1 ) - Please choose one Which of the following may not be an integral part of an object?

- ▶ State
- ▶ Behavior
- ▶ Protected data members (Page 12)
- ▶ All of given

Question No: 17 ( Marks: 1 ) - Please choose one Public methods of base class can-----be accessed in its derived class

- ▶ directly (Page 179) rep
- ▶ indirectly
- ▶ simultaneously
- ▶ non of the given

Question No: 18 ( Marks: 1 ) - Please choose one If a class D has been derived using protected inheritance from class B (If B is a protected base and D is derived class) then public and protected members of B----- accessed by member functions and friends of class D and classes derived from D



- ▶ can be [Click here for detail](#)
- ▶ cannot be
- ▶ does restrict to be
- ▶ not given

Question No: 19 ( Marks: 1 ) - Please choose one What is true about function templates?

- ▶ The compiler generates only one copy of the function template
- ▶ The compiler generates a copy of function respective to each type of data (Page 256)
- ▶ The compiler can only generate copy for the int type data
- ▶ None of the given.

Question No: 20 ( Marks: 1 ) - Please choose one Which of the following is an integral part of an object?

- ▶ State
- ▶ Behavior
- ▶ Unique identity
- ▶ All of the given (Page 12) rep 26

Question No: 21 ( Marks: 1 ) - Please choose one When the base class and the derived class have a member function with the same name, you must be more specific which function you want to call (using \_\_\_\_\_).

- ▶ scope resolution operator
- ▶ dot operator
- ▶ null operator
- ▶ Operator overloading [Click here for detail](#)

FINALTERM EXAMINATION Fall 2009 CS304- Object Oriented Programming (Session - 4)

Question No: 1 ( Marks: 1 ) - Please choose one A template provides a convenient way to make a family of

- ▶ variables and data members
- ▶ functions and classes (Object-Oriented Programming in C++)
- ▶ classes and exceptions
- ▶ programs and algorithms

Question No: 2 ( Marks: 1 ) - Please choose one Which one of the following terms must relate to polymorphism?

- ▶ Static allocation
- ▶ Static typing
- ▶ Dynamic binding (Page 239) rep
- ▶ Dynamic allocation

Question No: 3 ( Marks: 1 ) - Please choose one What is true about function templates?

- ▶ The compiler generates only one copy of the function template
- ▶ The compiler generates a copy of function respective to each type of data (Page 256) rep
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- ▶ None of the given.

Question No: 4 ( Marks: 1 ) - Please choose one Which of the following is the best approach if it is required to have more than one functions having exactly same functionality and implemented on different data types?

- ▶ Templates (Page 256) rep
- ▶ Overloading
- ▶ Data hiding
- ▶ Encapsulation 27

Question No: 5 ( Marks: 1 ) - Please choose one `template <> class Vector { }` This is an example of partial specialization.

- ▶ True (Page 287)
- ▶ False

Question No: 6 ( Marks: 1 ) - Please choose one Classes like `TwoDimensionalShape` and `ThreeDimensionalShape` would normally be concrete, while classes like `Sphere` and `Cube` would normally be abstract.

- ▶ True Click here for detail rep
- ▶ False

Question No: 7 ( Marks: 1 ) - Please choose one A non-virtual member function is defined in a base class and overridden in a derived class; if that function is called through a base-class pointer to a derived class object, the derived-class version is used.

- ▶ True
- ▶ False

Question No: 8 ( Marks: 1 ) - Please choose one Assume a class `Derv` that is privately derived from class `Base`. An object of class `Derv` located in `main()` can access

- ▶ public members of `Derv`. (Object-Oriented Programming in C++) rep
- ▶ protected members of `Derv`.
- ▶ private members of `Derv`.
- ▶ protected members of `Base`.

Question No: 9 ( Marks: 1 ) - Please choose one In order to define a class template, the first line of definition must be:

- ▶ `template` (Page 257) rep
- ▶ `typename`
- ▶ `Template Class`
- ▶ `Class 28`

Question No: 10 ( Marks: 1 ) - Please choose one If there is a pointer `p` to objects of a base class, and it contains the address of an object of a derived class, and both classes contain a nonvirtual member function, `ding()`, then the statement `p->ding();` will cause the version of `ding()` in the \_\_\_\_\_ class to be executed.

- ▶ `Base` (Object-Oriented Programming in C++)
- ▶ `Derived`
- ▶ `Abstract`
- ▶ `virtual`

Question No: 11 ( Marks: 1 ) - Please choose one When the base class and the derived class have a member function with the same name, you must be more specific which function you want to call (using \_\_\_\_\_).

- ▶ scope resolution operator
- ▶ dot operator
- ▶ null operator
- ▶ Operator overloading Click here for detail

Question No: 12 ( Marks: 1 ) - Please choose one Non Template Friend functions of a class are friends of \_\_\_\_\_ instance/s of that class.

- ▶ All Click here for detail
- ▶ One specific
- ▶ All instances of one date type
- ▶ None of the given options

Question No: 13 ( Marks: 1 ) - Please choose one The find() algorithm

- ▶ finds matching sequences of elements in two containers.
- ▶ finds a container that matches a specified container.
- ▶ takes iterators as its first two arguments. (Object-Oriented Programming in C++) rep
- ▶ takes container elements as its first two arguments.

Question No: 14 ( Marks: 1 ) - Please choose one If you define a vector v with the default constructor, and define another vector w with a one-argument constructor to a size of 11, and insert 3 elements into each of these vectors with `push_back()`, then the `size()` member function will return \_\_\_\_\_ for v and \_\_\_\_\_ for w.

- ▶ 11 for v and 3 for w.
- ▶ 0 for v and 0 for w.
- ▶ 0 for v and 3 for w.
- ▶ 3 for v and 11 for w. (Object-Oriented Programming in C++) rep 29

Question No: 15 ( Marks: 1 ) - Please choose one Which of the following may not be an integral part of an object?

- ▶ State
- ▶ Behavior
- ▶ Protected data members (Page 12)
- ▶ All of given

Question No: 16 ( Marks: 1 ) - Please choose one Which is not the Advantage of inheritance?

- ▶ providing class growth through natural selection. (Object-Oriented Programming in C++) rep
- ▶ facilitating class libraries.
- ▶ avoiding the rewriting of code
- ▶ providing a useful conceptual framework.

Question No: 17 ( Marks: 1 ) - Please choose one 1class DocElement { public: virtual void Print() { cout << "Generic element"; } }; 2class Heading : public DocElement { public: void Print() { cout << "Heading element"; } }; 3class Paragraph : public DocElement { public: void Print() { cout << "Paragraph element"; } }; void main() { DocElement \* p = new Paragraph(); p->Print(); } When you run this program, it will print out a single line to the console output. 30 What will be in that line?

Select one correct answer from the following list:

- ▶ Generic element
- ▶ Heading element
- ▶ Paragraph element
- ▶ Nothing will be printed.

Question No: 18 ( Marks: 1 ) - Please choose one When a virtual function is called by referencing a specific object by name and using the dot member selection operator (e.g., `squareObject.draw()`), the reference is resolved at compile time

- ▶ True
- ▶ False (Object-Oriented Programming in C++) (the function invocation is resolved at compile time)

Question No: 19 ( Marks: 1 ) - Please choose one In case of multiple inheritance a derived class inherits,

- ▶ Only the public member functions of its base classes
- ▶ Only the public data members of its base classes
- ▶ Both public data members and member functions of all its base classes Click here for detail
- ▶ Data members and member functions of any two base classes

Question No: 20 ( Marks: 1 ) - Please choose one When we write a class template the first line must be:

- ▶ template < class class\_name>
- ▶ template < class data\_type>
- ▶ template < class T > (Page 257) rep Here T can be replaced with any name but it is preferable.
- ▶ class class-name() class template

Question No: 21 ( Marks: 1 ) - Please choose one Which of the following is incorrect line regarding function template?

- ▶ template
- ▶ template
- ▶ Class (Page 257)
- ▶ template < class T, class U> 31

Question No: 22 ( Marks: 1 ) - Please choose one An STL container can not be used to,

- ▶ hold objects of class employee.
- ▶ store elements in a way that makes them quickly accessible.
- ▶ compile c++ programs. (Object-Oriented Programming in C++) rep
- ▶ organize the way objects are stored in memory

Question No: 23 ( Marks: 1 ) - Please choose one Algorithms can only be implemented using STL containers.

- ▶ True
- ▶ False (Object-Oriented Programming in C++) rep

Question No: 24 ( Marks: 1 ) - Please choose one Consider a class named Vehicle, which of the following can be the instance of class Vehicle? 1. Car 2. Computer 3. Desk 4. Ahmed 5. Bicycle 6. Truck

- ▶ 1, 4, 5
- ▶ 2, 5, 6
- ▶ 1, 2, 3, 6
- ▶ 1, 5, 6 (correct)

Question No: 25 ( Marks: 1 ) - Please choose one Consider the code below, class Fred { public: Fred(); ... }; int main() { Fred a[10]; Fred\* p = new Fred[10]; ... } Select the best option,

▶ Fred a[10]; calls the default constructor 09 times Fred\* p = new Fred[10]; calls the default constructor 10 times

- ▶ Produce an error 32
- ▶ Fred a[10]; calls the default constructor 11 times Fred\* p = new Fred[10]; calls the default constructor 11 times

▶ Fred a[10]; calls the default constructor 10 times Fred\* p = new Fred[10]; calls the default constructor 10 times Click here for detail

Question No: 26 ( Marks: 1 ) - Please choose one When a variable is define as static in a class then all object of this class,

- ▶ Have different copies of this variable
- ▶ Have same copy of this variable (Page 110)
- ▶ Can not access this variable
- ▶ None of given

Question No: 27 ( Marks: 1 ) - Please choose one The life of sub object is dependant on the life of master class in\_\_\_\_\_.

- ▶ Separation
- ▶ Composition (Page 53)
- ▶ Aggregation
- ▶ None of the given

Question No: 28 ( Marks: 1 ) - Please choose one \_\_\_\_\_, which means if A declares B as its friend it does NOT mean that A can access private data of B. It only means that B can access all data of A.

- ▶ Friendship is one way only Click here for detail
- ▶ Friendship is two way only
- ▶ NO Friendship between classes
- ▶ Any kind of friendship

Question No: 29 ( Marks: 1 ) - Please choose one Which of the following operators always takes no argument if overloaded?

- ▶ /
- ▶ -
- ▶ +
- ▶ ++ (Page 162)

Question No: 30 ( Marks: 1 ) - Please choose one In Private -----only member functions and friend classes or functions of a derived class can convert pointer or reference of derived object to that of parent object

- ▶ specialization
- ▶ inheritance (Page 216) rep
- ▶ abstraction
- ▶ composition

33 FINALTERM EXAMINATION Fall 2009 CS304- Object Oriented Programming (Session - 1)

Question No: 1 ( Marks: 1 ) - Please choose one Which one of the following terms must relate to polymorphism?

- ▶ Static allocation
- ▶ Static typing
- ▶ Dynamic binding (Page 239) rep
- ▶ Dynamic allocation

Question No: 2 ( Marks: 1 ) - Please choose one Multiple inheritance can be of type

- ▶ Public
- ▶ Private
- ▶ Protected
- ▶ All of the given Click here for detail

Question No: 3 ( Marks: 1 ) - Please choose one When a subclass specifies an alternative definition for an attribute or method of its superclass, it is \_\_\_\_\_ the definition in the superclass.

- ▶ overload
- ▶ overriding (Page 34)
- ▶ copy riding
- ▶ none of given

Question No: 4 ( Marks: 1 ) - Please choose one Like template functions, a class template may not handle all the types successfully.

- ▶ True (Page 258) rep
- ▶ False

Question No: 5 ( Marks: 1 ) - Please choose one It is sometimes useful to specify a class from which no objects will ever be created.

- ▶ True (Object-Oriented Programming in C++)
- ▶ False

Question No: 6 ( Marks: 1 ) - Please choose one Assume a class Derv that is privately derived from class Base. An object of class Derv located in main() can access

- ▶ public members of Derv. (Object-Oriented Programming in C++) rep
- ▶ protected members of Derv.
- ▶ private members of Derv.
- ▶ protected members of Base. 34

Question No: 7 ( Marks: 1 ) - Please choose one A pointer to a base class can point to objects of a derived class. \

- ▶ True Click here for detail
- ▶ False

Question No: 8 ( Marks: 1 ) - Please choose one A copy constructor is invoked when

- ▶ a function do not returns by value.
- ▶ an argument is passed by value. (Page 78) rep
- ▶ a function returns by reference.
- ▶ an argument is passed by reference.

Question No: 9 ( Marks: 1 ) - Please choose one A function call is resolved at run-time in \_\_\_\_\_

- ▶ non-virtual member function.
- ▶ virtual member function. (Page 239) rep
- ▶ Both non-virtual member and virtual member function.
- ▶ None of given

Question No: 10 ( Marks: 1 ) - Please choose one When the base class and the derived class have a member function with the same name, you must be more specific which function you want to call (using \_\_\_\_\_).

- ▶ scope resolution operator
- ▶ dot operator
- ▶ null operator
- ▶ Operator overloading Click here for detail

Question No: 11 ( Marks: 1 ) - Please choose one Each try block can have \_\_\_\_\_ no. of catch blocks.

- ▶ 1
- ▶ 2
- ▶ 3
- ▶ As many as necessary. Click here for detail rep

Question No: 12 ( Marks: 1 ) - Please choose one Two important STL associative containers are \_\_\_\_\_ and \_\_\_\_\_.

- ▶ set,map (Object-Oriented Programming in C++) rep
- ▶ sequence,mapping
- ▶ setmet,multipule
- ▶ sit,mat 35

Question No: 13 ( Marks: 1 ) - Please choose one The mechanism of selecting function at run time according to the nature of calling object is called,

- ▶ late binding
- ▶ static binding
- ▶ virtual binding
- ▶ None of the given options (Page 227) Dynamic binding means that target function for a call is selected at run time



Question No: 14 ( Marks: 1 ) - Please choose one An abstract class is useful when

- ▶ We do not derive any class from it.
- ▶ There are multiple paths from one derived class to another.
- ▶ We do not want to instantiate its object. (Object-Oriented Programming in C++) rep
- ▶ You want to defer the declaration of the class.

Question No: 15 ( Marks: 1 ) - Please choose one Which of the following is incorrect line regarding function template?

- ▶ template
- ▶ template
- ▶ Class (Page 257) rep
- ▶ template < class T, class U>

Question No: 16 ( Marks: 1 ) - Please choose one Which of the following is/are advantage[s] of generic programming?

- ▶ Reusability
- ▶ Writability
- ▶ Maintainability
- ▶ All of given (Page 256) rep

Question No: 17 ( Marks: 1 ) - Please choose one By default the vector data items are initialized to

- ▶ 0 Click here for detail rep
- ▶ 0.0
- ▶ 1
- ▶ null

Question No: 18 ( Marks: 1 ) - Please choose one Which one of the following functions returns the total number of elements in a vector.

- ▶ length();
- ▶ size(); (Page 318)
- ▶ ele();
- ▶ veclen();

36 Question No: 19 ( Marks: 1 ) - Please choose one Suppose you create an uninitialized vector as follows: vector evec; After adding the statment, evec.push\_back(21); what will happen?

- ▶ The following statement will add an element to the start (the back) of evec and will initialize it with the value 21.
- ▶ The following statement will add an element to the center of evec and will reinitialize it with the value 21.
- ▶ The following statement will delete an element to the end (the back) of evec and will reinitialize it with the value 21.
- ▶ The following statement will add an element to the end (the back) of evec and initialize it with the value 21. Click here for detail

Question No: 20 ( Marks: 1 ) - Please choose one An STL container can not be used to,

- ▶ hold objects of class employee.
- ▶ store elements in a way that makes them quickly accessible.
- ▶ compile c++ programs. (Object-Oriented Programming in C++) rep
- ▶ organize the way objects are stored in memory

Question No: 21 ( Marks: 1 ) - Please choose one Algorithms can only be implemented using STL containers.

- ▶ True

► False (Object-Oriented Programming in C++) rep

Question No: 22 ( Marks: 1 ) - Please choose one The main function of scope resolution operator (::) is,

- To define an object
- To define a data member
- To link the definition of an identifier to its declaration Click here for detail
- To make a class private

Question No: 23 ( Marks: 1 ) - Please choose one When is a constructor called?

- Each time the constructor identifier is used in a program statement
- During the instantiation of a new object (Object-Oriented Programming in C++)
- During the construction of a new class
- At the beginning of any program execution 37

Question No: 24 ( Marks: 1 ) - Please choose one Consider the code below, class Fred { public: Fred(); ... }; int main() { Fred a[10]; Fred\* p = new Fred[10]; ... } Select the best option,

- Fred a[10]; calls the default constructor 09 times Fred\* p = new Fred[10];
- Produce an error
- Fred a[10]; calls the default constructor 11 times Fred\* p = new Fred[10]; calls the default constructor 11 times
- Fred a[10]; calls the default constructor 10 times Fred\* p = new Fred[10]; calls the default constructor 10 times Click here for detail

Question No: 25 ( Marks: 1 ) - Please choose one Associativity can be changed in operator overloading.

- True
- False (Page 141)

Question No: 26 ( Marks: 1 ) - Please choose one A normal C++ operator that acts in special ways on newly defined data types is said to be

- glorified.
- encapsulated.
- classified.
- overloaded. (Object-Oriented Programming in C++)

Question No: 27 ( Marks: 1 ) - Please choose one Which operator can not be overloaded?

- The relation operator ( >= )
- Assignment operator ( = )
- Script operator ( [] )
- Conditional operator ( ? : ) (Page 141) 38

Question No: 28 ( Marks: 1 ) - Please choose one Suppose obj1 and obj2 are two objects of a user defined class A. An + operator is overloaded to add obj1 and obj2 using the function call obj1+obj2. Identify the correct function prototype against the given call?

- A operator + ( A &obj);
- int + operator(); (Page 143)
- int operator (plus) ();
- A operator(A &obj3);

Question No: 29 ( Marks: 1 ) - Please choose one Default constructor is such constructor which either has no -----or if it has some parameters these have----- values

- Parameter, temporary
- Null, Parameter
- Parameter, default (Page 75) rep

- ▶ non of the given

Question No: 30 ( Marks: 1 ) - Please choose one Public methods of base class can-----be accessed in its derived class

- ▶ directly (Page 179) rep
- ▶ inderectly
- ▶ simultaneously
- ▶ non of the given





